Covered Theory Proofs

Previously covered backbone results: canonical state, separator representation, injective minimality, expressive separations, predictive bridge, lower bounds, and identifiability

This note collects the proof surfaces that were already covered in the earlier proof-start drafts. Naming is kept deliberately restrained: the core structural results are typeset as theorems, while the two expressiveness adjuncts and the transferred lower-bound form are presented as direct statements with proofs.

Contents

1 Canonical predictive state **1**

2 Finite-horizon separator representation **2**

3 Injective minimality and the predictive skeleton **3**

4 Graph-traversal simulation of bounded deterministic computation **4**

5 Adaptive multi-hop traversal strictly exceeds one-shot retrieval **4**

6 Predictive closure exactness **5**

7 Budgeted topology necessity **6**

8 Topology recovery hardness and predictive-skeleton identifiability **7**

1. Canonical predictive state

Let H_t be the history random variable and F_t the future random variable, both taking values in standard Borel spaces. Write

$$\kappa_t(h) := \mathbb{P}(F_t \in \cdot \mid H_t = h).$$

Define predictive equivalence by

$$h \sim_t h' \iff \kappa_t(h) = \kappa_t(h'),$$

and let the canonical predictive realization be

$$\widetilde{m}_t^\star(h) := \kappa_t(h).$$

Call a measurable summary $S_t : \mathcal{H}_t \to \mathcal{S}_t$ exact when there exists a kernel Q_t such that

$$\kappa_t(h) = Q_t(S_t(h)) \qquad \text{for } \mathbb{P}_{H_t}\text{-a.e. } h.$$

Theorem. *The canonical predictive state is exact and minimal. More precisely:*

(i) for every measurable A,

$$\mathbb{P}(F_t \in A \mid H_t) = \mathbb{P}(F_t \in A \mid \widetilde{m}_t^\star(H_t)) \qquad a.s.;$$

(ii) if S_t is any exact summary, then there exists a measurable map f_t such that

$$\widetilde{m}_t^\star(h) = f_t(S_t(h)) \qquad \text{for } \mathbb{P}_{H_t}\text{-a.e. } h.$$

Hence every exact summary factors through the canonical predictive state.

Proof. For part (i), let A be measurable and define the evaluation functional

$$e_A(\mu) := \mu(A).$$

Then

$$e_A(\widetilde{m}_t^\star(H_t)) = e_A(\kappa_t(H_t)) = \kappa_t(H_t)(A) = \mathbb{P}(F_t \in A \mid H_t) \qquad \text{a.s.}$$

The right-hand side is therefore $\sigma(\widetilde{m}_t^\star(H_t))$-measurable and has the defining property of the conditional expectation of $\mathbf{1}_{\{F_t \in A\}}$ given $\widetilde{m}_t^\star(H_t)$. Thus

$$\mathbb{P}(F_t \in A \mid \widetilde{m}_t^\star(H_t)) = \mathbb{P}(F_t \in A \mid H_t) \qquad \text{a.s.}$$

For part (ii), if S_t is exact then by definition there is a kernel Q_t with

$$\kappa_t(h) = Q_t(S_t(h)) \qquad \text{a.e.}$$

Since $\widetilde{m}_t^\star(h) = \kappa_t(h)$, taking $f_t := Q_t$ yields

$$\widetilde{m}_t^\star(h) = f_t(S_t(h)) \qquad \text{a.e.}$$

This is the desired factorization. □

2. Finite-horizon separator representation

Fix a time t and a finite horizon $L \geq 1$, and write

$$F_t^{(L)} := X_{t+1:t+L}.$$

Assume that, conditional on $H_t = h$, the relevant variables $Y_{t,L}$ admit a finite-alphabet factorization on a tree decomposition of width at most w:

$$p_{t,L}^h(y) = \frac{1}{Z_{t,L}(h)} \prod_{c \in \mathcal{C}_{t,L}} \Phi_c^h(y_{C_c}).$$

Let $T_{t,L}^{\text{fut}}$ be the minimal connected subtree containing the future block and the retained observable interface. For each boundary edge $e = (a \to b)$ entering this future-support subtree, let

$$S_e := C_a \cap C_b, \qquad m_e^h(x_{S_e}) := \sum_{x_{V_e^{\text{out}} \setminus S_e}} \prod_{c \in T_e^{\text{out}}} \Phi_c^h(x_{C_c}),$$

and define the separator summary

$$M_t^{(L)}(h) := (I_t(h), (m_e^h)_{e \in \partial T_{t,L}^{\text{fut}}}).$$

Assume that the dependence of the inside factors on history is carried only by the retained interface assignment $I_t(h)$.

Theorem. *There exists a measurable map $g_{t,L}$ such that*

$$\kappa_{t,L}(h) := \mathbb{P}(F_t^{(L)} \in \cdot \mid H_t = h) = g_{t,L}(M_t^{(L)}(h)).$$

Equivalently, equal separator summaries imply equal finite-horizon predictive laws.

Proof. The running-intersection property implies that each outside branch T_e^{out} shares variables with the future-support subtree only through its separator S_e. Hence marginalizing the outside variables of branch e contributes precisely the function $m_e^h(x_{S_e})$.

Let

$$V^{\text{fut}} := \bigcup_{c \in T_{t,L}^{\text{fut}}} C_c.$$

After eliminating all outside variables, the unnormalized law on V^{fut} becomes

$$\tilde{p}_{t,L}^h(x_{V^{\text{fut}}}) = \Phi_{\text{fut}}^{I_t(h)}(x_{V^{\text{fut}}}) \cdot \prod_{e \in \partial T_{t,L}^{\text{fut}}} m_e^h(x_{S_e}),$$

where the inside factor $\Phi_{\text{fut}}^{I_t(h)}$ depends on h only through $I_t(h)$ by assumption. Now sum out the latent variables inside the future-support subtree but outside the actual future block. The resulting unnormalized law of $F_t^{(L)}$ depends on h only through

$$(I_t(h), (m_e^h)_e) = M_t^{(L)}(h).$$

Normalizing this law yields the claimed measurable map $g_{t,L}$. □

3. Injective minimality and the predictive skeleton

Let $\bar{M}_t^{(L)}(h) = (I_t(h), (\bar{m}_e^h)_e)$ denote the normalized separator tuple, and let

$$U_t^{(L)}(h) := \bigotimes_{e \in \partial T_{t,L}^{\text{fut}}} \bar{m}_e^h$$

be its normalized product tensor. Write

$$\Gamma_{t,L}(i, U)$$

for the normalized future law induced by a reachable canonical boundary state (i, U).

Say that the process class is separator-faithful at (t, L) when the map $(i, U) \mapsto \Gamma_{t,L}(i, U)$ is injective on the reachable boundary-state set.

Theorem. *Under separator-faithfulness,*

$$\kappa_{t,L}(h) = \kappa_{t,L}(h') \iff \bar{M}_t^{(L)}(h) = \bar{M}_t^{(L)}(h')$$

for all reachable histories h, h'. Consequently the normalized separator tuple is an exact normal form of the finite-horizon predictive quotient.

Proof. The implication

$$\bar{M}_t^{(L)}(h) = \bar{M}_t^{(L)}(h') \Longrightarrow \kappa_{t,L}(h) = \kappa_{t,L}(h')$$

is exactly the separator-representation theorem of the previous section.

For the converse, assume

$$\kappa_{t,L}(h) = \kappa_{t,L}(h').$$

By construction this means

$$\Gamma_{t,L}(I_t(h), U_t^{(L)}(h)) = \Gamma_{t,L}(I_t(h'), U_t^{(L)}(h')).$$

Separator-faithfulness forces equality of the two reachable boundary states:

$$(I_t(h), U_t^{(L)}(h)) = (I_t(h'), U_t^{(L)}(h')).$$

The equality of the product tensors implies equality of each normalized branch message, since the tensor is a direct product of normalized components. Hence

$$\bar{m}_e^h = \bar{m}_e^{h'} \qquad \forall e,$$

and therefore

$$\bar{M}_t^{(L)}(h) = \bar{M}_t^{(L)}(h').$$

□

Without faithfulness. Define predictive indistinguishability on reachable boundary states by

$$(i, U) \equiv_{t,L}^{\text{pred}} (i', U') \iff \Gamma_{t,L}(i, U) = \Gamma_{t,L}(i', U').$$

The quotient

$$\text{Skel}_{t,L} := \mathcal{C}_{t,L}^{\text{reach}} / \equiv_{t,L}^{\text{pred}}$$

is then the exact canonical object that remains when injectivity fails.

Proof. By definition, two reachable boundary states are equivalent exactly when they induce the same future law. Hence equal predictive laws correspond bijectively to equal quotient classes. That quotient is therefore the coarsest object that preserves exact prediction and the finest object that remains identifiable without faithfulness. □

4. Graph-traversal simulation of bounded deterministic computation

A graph-traversal transducer maintains a finite control state, a pointer to a current node of a persistent graph, local node labels, and a local transition rule that may rewrite the current node, move to a neighbor, or create one new blank node when the pointer attempts to step beyond the current boundary of the graph.

Statement. Every deterministic single-tape Turing machine M admits a graph-traversal transducer simulation with the following property: for every input x and every time t, the transducer configuration after t steps is the exact path encoding of the Turing-machine configuration after t steps. If M halts after $T(x)$ steps, the transducer halts with the same accept/reject outcome and uses at most

$$|x| + T(x) + 2$$

nodes.

Proof. Encode the tape as a doubly linked path whose node labels are the tape symbols, and place the pointer at the head position. Use the Turing-machine control state as the transducer control state. One transducer step now performs exactly the local Turing update: rewrite the current symbol, move left, right, or stay, and if the head moves beyond the realized path, create one fresh blank node and attach it to the boundary.

The initial transducer graph is the obvious path encoding of the input tape, so the simulation claim is true at time 0. If it is true at time t, then the local update rule reproduces exactly the Turing-machine tape rewrite, head move, and control-state update at time $t + 1$. This proves the inductive step. Halting correctness is immediate because halting states are copied into the transducer control state. The node bound follows because the input contributes $|x|$ nodes initially and each machine step can create at most one new boundary blank. □

5. Adaptive multi-hop traversal strictly exceeds one-shot retrieval

Fix integers $n \geq 2$ and $k \geq 1$. The layered pointer-chasing problem is specified by a start address $s \in [n]$, maps $f_1, \ldots, f_k : [n] \to [n]$, and a bit table $b : [n] \to \{0, 1\}$. Its value is

$$\mathrm{PC}_{n,k} := b(f_k \circ \cdots \circ f_1(s)).$$

An adaptive traversal machine may query one table entry at a time and choose the next query based on earlier answers. A one-shot retrieval machine must fix all queried table entries nonadaptively before seeing any answers.

Statement. The adaptive traversal model computes $\mathrm{PC}_{n,k}$ exactly with $k + 1$ reads, whereas every exact one-shot retrieval machine requires budget at least

$$B \geq kn + 1.$$

Hence, under matched small budgets, adaptive multi-hop traversal is strictly stronger than one-shot retrieval.

Proof. For the upper bound, read $f_1(s)$, then $f_2(f_1(s))$, and continue adaptively until the final address

$$\nu_k = f_k \circ \cdots \circ f_1(s)$$

is reached. One further query to $b(\nu_k)$ returns the exact answer. This uses exactly $k + 1$ reads.

For the lower bound, suppose a one-shot retrieval scheme is exact. First, the entry $(1, s)$ must be queried; otherwise two instances that differ only in the value of $f_1(s)$ but are identical on all queried entries would produce the same transcript and different outputs. Second, every address in every hidden layer $\ell = 2, \ldots, k$ must be queried. If some $a \in [n]$ at hidden layer ℓ were omitted, one could force the actual path to arrive at a, alter only the unqueried value $f_\ell(a)$, keep all queried entries fixed, and thereby change the output while keeping the transcript identical. Third, every final label $b(j)$ must be queried for the same reason. Therefore the scheme must query

$$1 + (k - 1)n + n = kn + 1$$

entries in total. □

6. Predictive closure exactness

Let a workspace be denoted by W, and write

$$\mathrm{Close}(W, W_c; H, b)$$

for a transition that replaces a history-side region H by an archived block b. For finite horizon L, let

$$\Xi_{t,L}(W, h) = \big(I_{t,L}(W, h), (m_e^{W,h})_e\big)$$

be the workspace-induced separator summary: retained interface plus exact boundary messages. Assume workspace-to-core realizability in the form

$$\mathrm{Pred}_{t,L}(W, h) = g_{t,L}^{\mathrm{core}(W,h)}(\Xi_{t,L}(W, h)).$$

Assume further that the close operation preserves the future-support core, replaces the branch H by its exact stored separator summary, leaves all other boundary messages unchanged, and preserves the retained interface.

Theorem. *Under those assumptions,*

$$\mathrm{Pred}_{t,L}(W_c, h) = \mathrm{Pred}_{t,L}(W, h).$$

That is, a predictively safe close preserves the finite-horizon predictive law exactly.

Proof. By workspace-to-core realizability,

$$\mathrm{Pred}_{t,L}(W, h) = g_{t,L}^{\mathrm{core}(W,h)}(\Xi_{t,L}(W, h)), \qquad \mathrm{Pred}_{t,L}(W_c, h) = g_{t,L}^{\mathrm{core}(W_c,h)}(\Xi_{t,L}(W_c, h)).$$

The assumptions imply that the future-support core is unchanged, so the two core maps agree, and that the full separator summary is unchanged:

$$\Xi_{t,L}(W_c, h) = \Xi_{t,L}(W, h).$$

Substituting these equalities into the displayed representations gives

$$\mathrm{Pred}_{t,L}(W_c, h) = \mathrm{Pred}_{t,L}(W, h).$$

□

Full-future extension. If every block stores a projectively consistent all-horizon family of exact finite-horizon summaries, then the full future law is also preserved.

Proof. The theorem gives equality of every finite-horizon predictive law. Cylinder-event probabilities therefore agree at every horizon, and projective consistency implies equality of the induced full-future measure by cylinder uniqueness. □

7. Budgeted topology necessity

Fix $m \geq 2$ and let

$$k := \lceil \log_2 m \rceil.$$

In one episode sample

$$J \sim \mathrm{Unif}([m]), \qquad V \sim \mathrm{Unif}(\{0,1\}^k),$$

independently, and define the object table by

$$U_i := 0^k \ (i \neq J), \qquad U_J := V.$$

The prefix reveals all objects (i, U_i), then a topology token $q \to J$, then a filler block longer than the local window, then the query token q. The answer block is

$$Y_{1:k} := V.$$

A topology-aware predictor in $\mathcal{C}_{\mathrm{top}}$ may retain the explicit separator address from q to the active object. A topology-free predictor in $\mathcal{C}_{\mathrm{set}}(b, r)$ may keep the payload table, but its only extra resources are a b-bit controller state

$$C \in \{0,1\}^b$$

formed after the setup and an adaptive query-time read budget of at most r payload reads.

Theorem. *For the one-active-separator indexing process:*

(i) there exists a topology-aware predictor in $\mathcal{C}_{\mathrm{top}}$ with one-episode answer-block log-loss 0 *using $O(1)$ query-time reads;*

(ii) every topology-free predictor $\pi \in \mathcal{C}_{\mathrm{set}}(b, r)$ satisfies

$$\mathbb{E}[L_\pi^{\mathrm{ans}}] \geq \left(1 - \frac{2^b r}{m}\right)_+ (k - b)_+,$$

with base-2 log-loss.

Proof. For part (i), store the object table and the explicit address $q \to J$. At query time, read that address to recover J, then read U_J. Since $U_J = V = Y_{1:k}$, the answer block is predicted with probability 1 and the log-loss is zero.

For part (ii), fix $\pi \in \mathcal{C}_{\mathrm{set}}(b, r)$. For each controller state c, run the query-time protocol under the hypothetical transcript in which every read payload is 0^k, and let $A_\pi(c) \subseteq [m]$ be the set of indices visited in that zero transcript. Then

$$|A_\pi(c)| \leq r.$$

Define the global coverage set

$$\mathcal{U}_\pi := \bigcup_{c \in \{0,1\}^b} A_\pi(c),$$

so

$$|\mathcal{U}_\pi| \leq 2^b r.$$

Let

$$E_\pi := \{J \notin \mathcal{U}_\pi\}$$

be the miss event. Since J is uniform on $[m]$,

$$\mathbb{P}(E_\pi) \geq 1 - \frac{2^b r}{m}.$$

On E_π, the active object is never read. Indeed, before the active object is found, every observed payload equals 0^k, so the actual adaptive execution coincides with the zero-transcript execution determined by the controller state. Because $J \notin \mathcal{U}_\pi$, that execution never queries object J. Hence the full query transcript T_π^{qry} is a deterministic function of C on E_π.

Now E_π depends only on J and the fixed predictor, so it is independent of V. Therefore

$$H(V \mid E_\pi) = H(V) = k.$$

Since T_π^{qry} is a function of C on E_π,

$$I(V; T_\pi^{\mathrm{qry}} \mid E_\pi) \leq I(V; C \mid E_\pi) \leq H(C \mid E_\pi) \leq b.$$

Thus

$$H(V \mid T_\pi^{\mathrm{qry}}, E_\pi) \geq k - b.$$

Under sequential log-loss, the expected answer-block loss conditional on the transcript is at least the posterior entropy, so

$$\mathbb{E}[L_\pi^{\mathrm{ans}} \mid E_\pi] \geq H(V \mid T_\pi^{\mathrm{qry}}, E_\pi) \geq k - b.$$

Multiplying by the miss-event probability gives

$$\mathbb{E}[L_\pi^{\mathrm{ans}}] \geq \mathbb{P}(E_\pi)(k - b) \geq \left(1 - \frac{2^b r}{m}\right)(k - b).$$

Writing the bound with positive parts makes it valid in every parameter regime. □

8. Topology recovery hardness and predictive-skeleton identifiability

Let $\mathcal{G}$ be a family of latent graphs. Each $G \in \mathcal{G}$ induces a token-sequence law P_G, and the learner observes only

$$D_n \sim P_G^{\otimes n}.$$

Define predictive equivalence by

$$G \equiv_{\text{pred}} G' \iff P_G(X_{t+1:\infty} \in A \mid H_t = h) = P_{G'}(X_{t+1:\infty} \in A \mid H_t = h)$$

for every admissible (t, h, A), and define the predictive skeleton by

$$\mathrm{Sk}(G) := [G]_{\equiv_{\text{pred}}}.$$

Theorem. *Two conclusions follow.*

(i) If $G \neq G'$ but $G \equiv_{\text{pred}} G'$, then no estimator based only on token data can recover the exact graph uniformly: for every estimator $\widehat{G}_n$,

$$\max\Big\{\mathbb{P}_G[\widehat{G}_n \neq G], \mathbb{P}_{G'}[\widehat{G}_n \neq G']\Big\} \geq \frac{1}{2}.$$

(ii) Every graph functional F that is identifiable from token data factors through the predictive skeleton. Equivalently, there exists a map ϕ such that

$$F = \phi \circ \mathrm{Sk}\,.$$

Proof. For part (i), if $G \equiv_{\text{pred}} G'$, then the observed data have the same law under both graphs:

$$D_n \sim P_G^{\otimes n} = P_{G'}^{\otimes n}.$$

Call this common law Q_n. Then

$$\mathbb{P}_G[\widehat{G}_n \neq G] = Q_n[\widehat{G}_n \neq G], \qquad \mathbb{P}_{G'}[\widehat{G}_n \neq G'] = Q_n[\widehat{G}_n \neq G'].$$

Because $G \neq G'$, one output cannot be correct for both worlds simultaneously, so

$$\mathbf{1}\{\widehat{G}_n \neq G\} + \mathbf{1}\{\widehat{G}_n \neq G'\} \geq 1$$

pointwise. Taking expectations under Q_n yields the lower bound $1/2$.

For part (ii), let F be identifiable from token data. If there existed $G \equiv_{\text{pred}} G'$ with

$$F(G) \neq F(G'),$$

then the same identical-data-law argument would show that no estimator could consistently distinguish the two values. Therefore

$$G \equiv_{\text{pred}} G' \implies F(G) = F(G').$$

So F is constant on predictive-equivalence classes and hence factors through the quotient map $G \mapsto \mathrm{Sk}(G)$. □

Transferred finite-sample lower bounds. Suppose a direct graphical-model family $\{P_\Theta^Z : \Theta \in \Xi\}$ has support graph $G(\Theta)$, and let

$$E : \mathcal{Z}^p \to \mathcal{X}^\ell$$

be an injective block encoder. Any token-only topology estimator on the encoded corpus induces a direct support estimator by composition with E.

Proof. Given a direct-sample estimator problem, define

$$\widehat{G}_n^{\mathrm{dir}}(D_n^{\mathrm{dir}}) := \widehat{G}_n^{\mathrm{NTP}}(E(D_n^{\mathrm{dir}})).$$

If the token-only estimator achieved smaller minimax error than the direct estimator, the composed rule would contradict the direct lower bound. Thus every direct support-recovery lower bound transfers to the corresponding token-only topology-recovery problem. □